Group 5 Analysis

Group 05

1 Introduction

This project explores which household-related variables contribute to the number of people living in a household using Dataset05 from the Family Income and Expenditure Survey (FIES) in the Philippines. The response variable is `Total.Number.of.Family.members`, which is a count outcome. Therefore, Poisson and Negative Binomial models are appropriate candidate models.

We begin with exploratory data analysis, then fit a Poisson regression model as the baseline count model. Considering count data often exhibit overdispersion, a Negative Binomial model is also fitted for comparison to explore whether it provides a better fit. The research aims to identify the main household characteristics associated with household size and interpret these relationships carefully.

```r
library(tidyverse)
library(janitor)
library(skimr)
library(GGally)
library(MASS)
library(AER)
library(car)
library(broom)
library(performance)
library(pscl)
library(knitr)
library(kableExtra)
library(ggplot2)
library(patchwork)
```

2 Data Description and Processing

2.1 Data Overview

```
Rows: 1,788
Columns: 11
$ Total.Household.Income         <dbl> 192922, 161843, 535899, 312579, 154715,~
$ Region                         <chr> "IX - Zasmboanga Peninsula", "IX - Zasm~
$ Total.Food.Expenditure         <dbl> 114258, 78176, 92464, 133445, 39580, 58~
$ Household.Head.Sex             <chr> "Male", "Male", "Male", "Male", "Male",~
$ Household.Head.Age             <dbl> 56, 66, 46, 46, 46, 40, 57, 61, 45, 32,~
$ Type.of.Household              <chr> "Single Family", "Extended Family", "Si~
$ Total.Number.of.Family.members <dbl> 3, 6, 4, 14, 3, 6, 3, 5, 8, 5, 1, 6, 4,~
$ House.Floor.Area               <dbl> 32, 24, 12, 49, 32, 20, 35, 12, 12, 11,~
$ House.Age                      <dbl> 28, 6, 3, 21, 1, 2, 39, 4, 4, 5, 46, 15~
$ Number.of.bedrooms             <dbl> 2, 0, 1, 3, 2, 0, 2, 1, 1, 1, 3, 1, 1, ~
$ Electricity                    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
```

```
 Total.Household.Income    Region          Total.Food.Expenditure
 Min.   :  19730        Length:1788        Min.   :  5408
 1st Qu.:  85264        Class :character   1st Qu.: 40500
 Median : 126567        Mode  :character   Median : 59256
 Mean   : 191001                           Mean   : 69645
 3rd Qu.: 205489                           3rd Qu.: 87221
 Max.   :6071030                           Max.   :434881
 Household.Head.Sex Household.Head.Age Type.of.Household
 Length:1788        Min.   :16.00      Length:1788
 Class :character   1st Qu.:41.00      Class :character
 Mode  :character   Median :50.00      Mode  :character
                    Mean   :50.95
                    3rd Qu.:61.00
                    Max.   :98.00
 Total.Number.of.Family.members House.Floor.Area   House.Age
 Min.   : 1.000                 Min.   :  7.00   Min.   : 0
 1st Qu.: 3.000                 1st Qu.: 18.00   1st Qu.: 7
 Median : 4.000                 Median : 28.00   Median :15
 Mean   : 4.552                 Mean   : 38.42   Mean   :16
 3rd Qu.: 6.000                 3rd Qu.: 45.00   3rd Qu.:20
 Max.   :15.000                 Max.   :638.00   Max.   :93
 Number.of.bedrooms  Electricity
 Min.   :0.000      Min.   :0.0000
 1st Qu.:1.000      1st Qu.:1.0000
```

```
 Median :2.000      Median :1.0000
 Mean   :1.732      Mean   :0.8082
 3rd Qu.:2.000      3rd Qu.:1.0000
 Max.   :6.000      Max.   :1.0000
```

The dataset contains 1,788 household-level observations and 11 variables. These variables describe household income, expenditure, demographic characteristics of the household head, housing conditions, and household size. Both numeric and categorical variables are included in the dataset, as identified during the initial inspection using the `glimpse()` and `summary()`. In particular, `Total.Number.of.Family.members` is the response variable of interest, while `Region`, `Household.Head.Sex`, and `Type.of.Household` are categorical variables, and the remaining variables are numeric.

2.2 Initial Checks

```r
dim(df)
```

```
[1] 1788   11
```

```r
names(df)
```

```
 [1] "Total.Household.Income"         "Region"
 [3] "Total.Food.Expenditure"         "Household.Head.Sex"
 [5] "Household.Head.Age"             "Type.of.Household"
 [7] "Total.Number.of.Family.members" "House.Floor.Area"
 [9] "House.Age"                      "Number.of.bedrooms"
[11] "Electricity"
```

```r
colSums(is.na(df))
```

```
        Total.Household.Income                         Region
                             0                              0
        Total.Food.Expenditure             Household.Head.Sex
                             0                              0
            Household.Head.Age               Type.of.Household
                             0                              0
Total.Number.of.Family.members                House.Floor.Area
                             0                              0
                     House.Age             Number.of.bedrooms
```

```
                            0                                0
                  Electricity
                            0
```

2.3 Check Unique Values for Categorical Variables

```
# A tibble: 1 x 2
  Region                       n
  <chr>                    <int>
1 IX - Zasmboanga Peninsula  1788
```

```
# A tibble: 2 x 2
  Household.Head.Sex     n
  <chr>              <int>
1 Male                1434
2 Female               354
```

```
# A tibble: 3 x 2
  Type.of.Household                           n
  <chr>                                   <int>
1 Single Family                            1254
2 Extended Family                           522
3 Two or More Nonrelated Persons/Members     12
```

```
# A tibble: 2 x 2
  Electricity     n
        <dbl> <int>
1           1  1445
2           0   343
```

2.4 Variable Conversion and Transformed Variables

```r
df_clean <- df %>%
  mutate(
    Region = as.factor(Region),
    Household.Head.Sex = as.factor(Household.Head.Sex),
    Type.of.Household = as.factor(Type.of.Household),
    Electricity = factor(Electricity, levels = c(0, 1), labels = c("No", "Yes"))
  )
```

```r
df_clean <- df_clean %>%
  mutate(
    log_income = log(Total.Household.Income),
    log_food_exp = log(Total.Food.Expenditure),
    log_floor_area = log(House.Floor.Area)
  )
```

Initial checks indicate that there are no missing values in the dataset, therefore no imputation or complete-case filtering was required, and the full sample of 1,788 households was retained throughout the analysis. Frequency tables for the categorical variables show that all observations come from a single region, IX - Zamboanga Peninsula. Consequently, `Region` contains no within-sample variation and cannot contribute to explaining differences in household size across households. It is therefore excluded from the regression analysis.

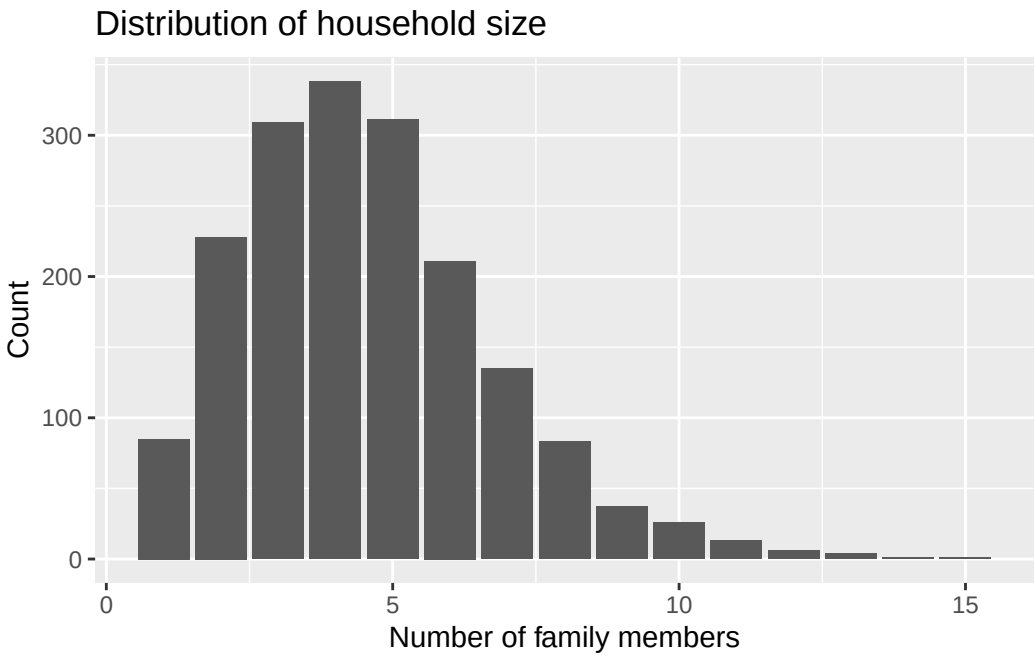
The categorical variables `Household.Head.Sex`, `Type.of.Household`, and `Electricity` were converted into factor variables before modelling. This ensures that they are treated appropriately as categorical predictors in the GLM framework. Moreover, three skewed numeric variables - household income, food expenditure, and floor area - were log-transformed. These transformations reduce right-skewness, improve comparability across households and make the relationship between predictors and household size easier to interpret.

3 Exploratory Data Analysis

3.1 Response Variable Distribution

Table 1: Summary Statistics of Family Size

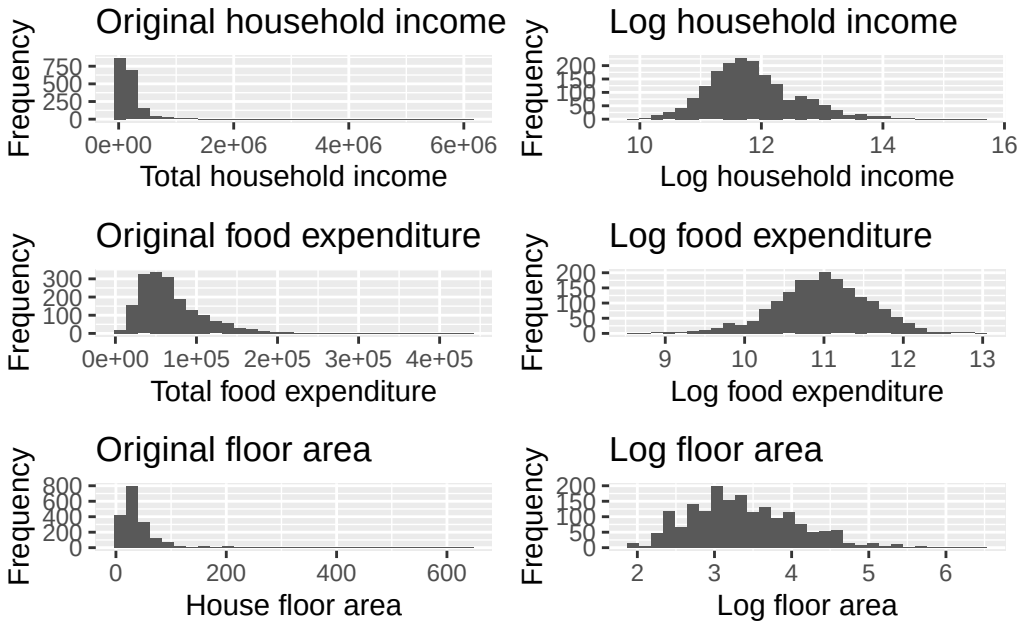
mean_family_size	var_family_size	min_family_size	max_family_size
4.55	4.79	1	15



We use `Total.Number.of.Family.members` as the response variable. It ranges from 1 to 15 and is clearly a discrete count variable. The distribution of household size appears to be moderately right-skewed, with most households containing between 3 and 6 members. This pattern is consistent with the nature of household survey data, where medium-sized households are common and very large households are relatively rare.

The sample mean of household size is 4.55, while the sample variance is 4.79. Because the sample variance is close to the sample mean, there is no strong descriptive indication of substantial overdispersion at this stage. This supports the use of a Poisson regression model as a reasonable starting point for the count-data analysis.

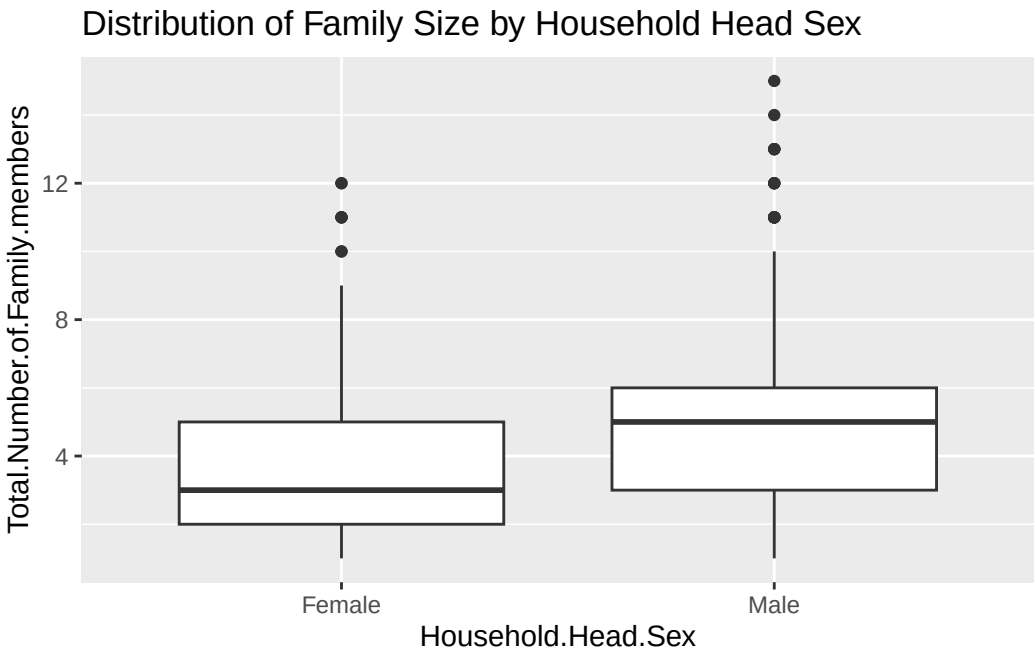
3.2 Numeric Variable Distributions



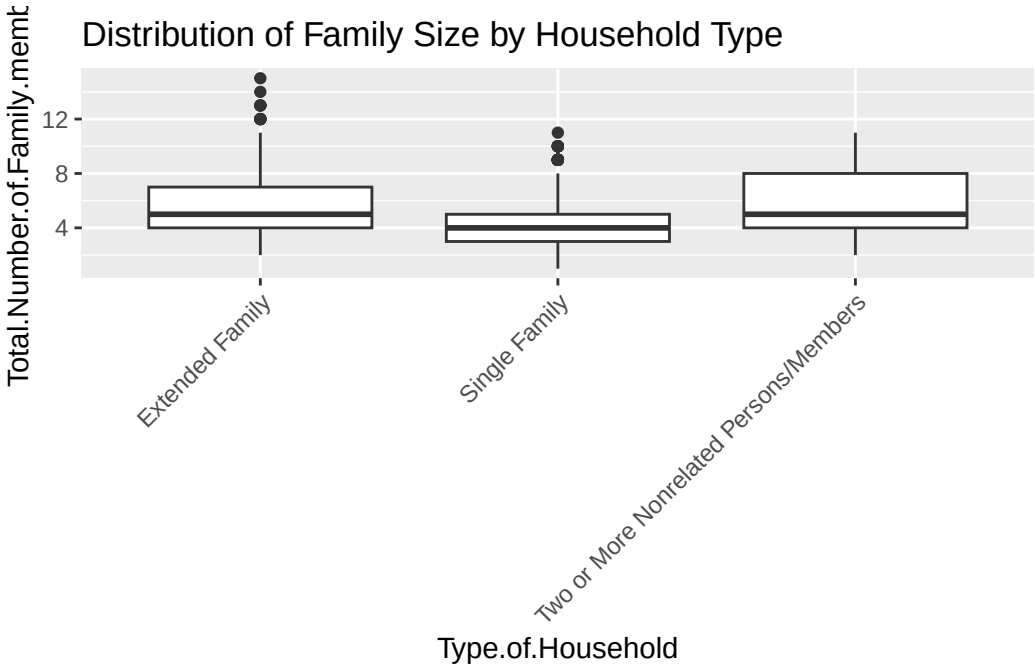
The distributions of `Total.Household.Income`, `Total.Food.Expenditure`, and `House.Floor.Area` are clearly right-skewed in original scales. For each of these variables, the raw histograms show that most observations are concentrated at relatively low values, with only a few households taking much larger values. This indicates the presence of long right tails and suggests that the original scales may not be ideal for modelling.

After log transformation, the distributions of these variables become more compact and regular. The transformed histograms show reduced skewness and a clearer concentration around central values. This provides visual support for the use of `log_income`, `log_food_exp`, and `log_floor_area` in the regression models. Overall, the transformation improves both model adequacy and interpretability, especially when comparing households with widely different economic and housing conditions.

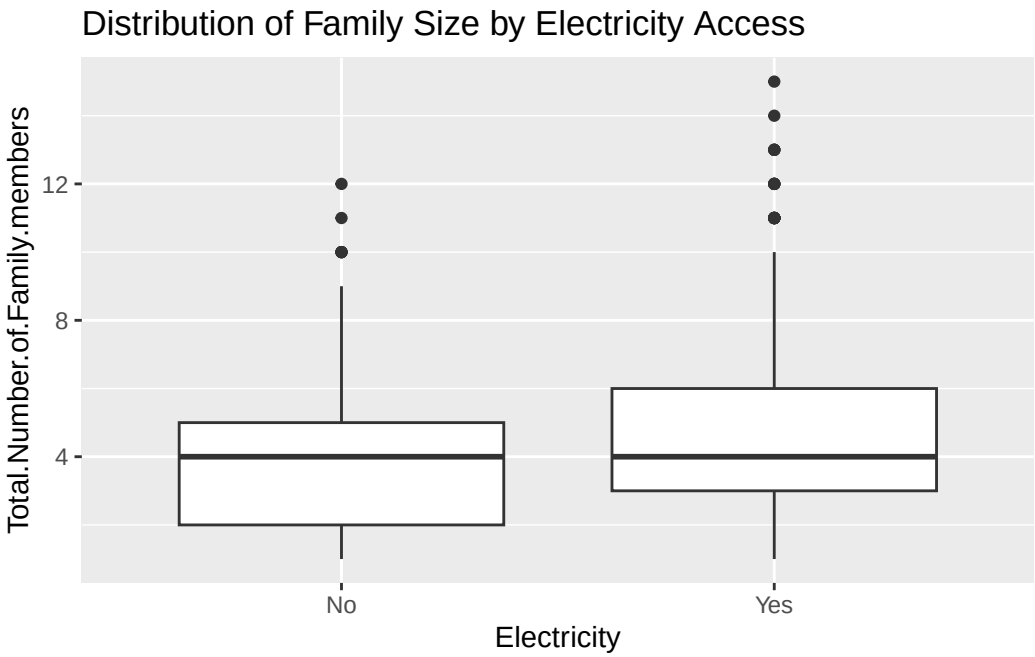
3.3 Household Size by Categorical Predictors



The boxplot shows that male-headed households tend to have larger household size than female-headed households. The median household size is higher for male-headed households, and their distribution also has a wider upper tail, indicating that larger households are more common in this group.

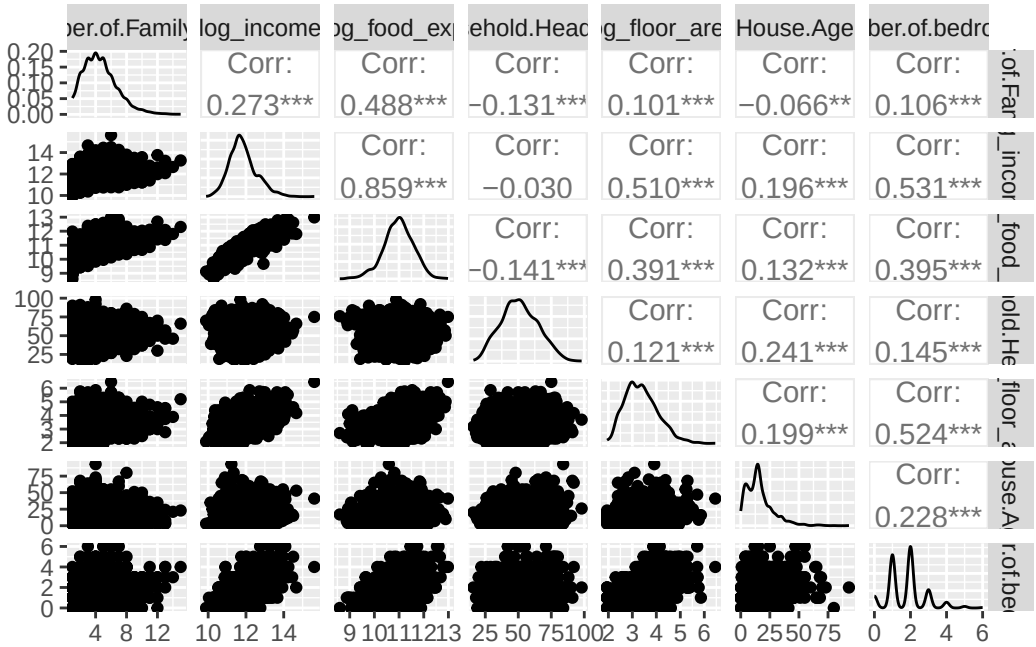


The results suggest that there are differences in household size across household types. Single-family households appear to have the smallest and most concentrated household size distribution, with a median of around 4. By contrast, extended-family households have a higher median and a wider upper tail, indicating that larger households are more common in this group. The category of two or more unrelated members also tends to have relatively large household size, although this group is very small and results should be interpreted with caution.



The results suggest that the relationship between electricity access and household size is weaker than that observed for household type or sex of household head. The median household size is similar across the two groups, but households with electricity show a wider upper range and more large-household outliers. This indicates that electricity access may still be associated with household size, although the pattern appears less evident than for the other categorical predictors.

3.4 Household Size Against Numeric Predictors



The pairwise plot was used to summarise the relationships among the numeric variables. In relation to household size, the strongest positive association appears with `log_food_exp`, followed by a weaker positive association with `log_income`. Household size is also positively related, though more weakly, to `log_floor_area` and `Number.of.bedrooms`. These patterns are plausible, since larger households may require more food expenditure and may also occupy larger dwellings with more bedrooms.

In contrast, the association between household size and `House.Age` is very weak, while the relationship with `Household.Head.Age` is slightly negative. The latter may reflect household life-cycle effects, where households become smaller at older ages as children leave the home. The pairwise plot also reveals substantial correlation among some explanatory variables, especially between `log_income` and `log_food_exp`, and between `log_floor_area` and `Number.of.bedrooms`. These results suggest that some predictors may contain overlapping information, so multicollinearity should be considered during model fitting.

3.5 Implications of EDA for Modelling

Overall, the exploratory analysis suggests that a count-data modelling approach is appropriate. Household size is not only a discrete response variable, but the distribution is moderately right-skewed and the mean is close to the variance, so Poisson regression provides a reasonable starting point. The descriptive results indicate that household type, the sex of the household

head, and various economic and housing variables may serve as significant predictors. Some numeric variables also exhibit strong skewness, which supports the use of log transformations. Additionally, the correlation between some predictors suggests that multicollinearity should be checked during modelling.

4 GLM Modelling and Results

4.1 Model Specification

In this study, the response variable is household size, defined as the number of members in each household. Since household size is a non-negative integer count observed for each household, a Poisson regression model was adopted as the baseline specification.

This analysis uses the simple Poisson case rather than a Poisson model with an offset, because the response is a direct count per household rather than an event count observed over varying exposure times, population sizes, or risk periods.

Let Y_i denote the household size for household i, and let X_i be the vector of explanatory variables. The Poisson regression model is specified as

$$Y_i \mid X_i \sim \text{Poisson}(\mu_i), \qquad i = 1, \dots, n,$$

with log link function

$$\log(\mu_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip}.$$

Under the Poisson assumption,

$$\mathbb{E}(Y_i \mid X_i) = \text{Var}(Y_i \mid X_i) = \mu_i.$$

An overdispersion check was then conducted after fitting the Poisson model. A Negative Binomial model was also fitted as a more flexible alternative specification for comparison.

4.2 Poisson Regression Results

```
Call:
glm(formula = Total.Number.of.Family.members ~ log_income + log_food_exp +
    Household.Head.Age + log_floor_area + House.Age + Number.of.bedrooms +
    Household.Head.Sex + Type.of.Household + Electricity, family = poisson(link = "log"),
    data = df_clean)

Coefficients:
                                                        Estimate Std. Error
(Intercept)                                            -2.1341812  0.2579909
log_income                                             -0.3803531  0.0369108
log_food_exp                                            0.7577621  0.0417800
Household.Head.Age                                     -0.0023998  0.0009318
log_floor_area                                          0.0076785  0.0200140
House.Age                                              -0.0032621  0.0010749
Number.of.bedrooms                                      0.0053244  0.0142696
Household.Head.SexMale                                  0.2015358  0.0314002
Type.of.HouseholdSingle Family                         -0.2923877  0.0254667
Type.of.HouseholdTwo or More Nonrelated Persons/Members  0.0128239  0.1207469
ElectricityYes                                         -0.0389573  0.0324903
                                                       z value Pr(>|z|)
(Intercept)                                             -8.272  < 2e-16 ***
log_income                                             -10.305  < 2e-16 ***
log_food_exp                                            18.137  < 2e-16 ***
Household.Head.Age                                      -2.575  0.01001 *
log_floor_area                                           0.384  0.70123
House.Age                                               -3.035  0.00241 **
Number.of.bedrooms                                       0.373  0.70905
Household.Head.SexMale                                   6.418 1.38e-10 ***
Type.of.HouseholdSingle Family                         -11.481  < 2e-16 ***
Type.of.HouseholdTwo or More Nonrelated Persons/Members   0.106  0.91542
ElectricityYes                                          -1.199  0.23051
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1854.6  on 1787  degrees of freedom
Residual deviance: 1040.0  on 1777  degrees of freedom
AIC: 6917.3
```

```
Number of Fisher Scoring iterations: 4
```

A Poisson regression model was fitted to examine the relationship between household size and socioeconomic characteristics. The model showed a substantial reduction in deviance, indicating good model fit. `log_food_exp` was strongly positively associated with family size, specifically, a one-unit increase in log food expenditure is associated with an expected household size that is $e^{0.758} = 2.133$ times as large. In contrast, `log_income` showed a negative relationship, suggesting that higher-income households tend to be smaller, with the expected household size changing by a factor of $e^{-0.380} = 0.684$. Household structure variables, particularly single-family households, were also significant predictors. In contrast, variables such as floor area, number of bedrooms, and electricity access were not statistically significant.

4.3 Check for Overdispersion

```r
deviance(model_poisson) / df.residual(model_poisson)
```

```
[1] 0.5852337
```

Overdispersion was assessed using the ratio of residual deviance to residual degrees of freedom. The estimated value was

$$\frac{\text{Residual deviance}}{\text{Residual df}} = \frac{1040.0}{1777} = 0.585.$$

The value is below 1, indicating slight underdispersion rather than overdispersion. However, the deviation is not substantial, and the Poisson model remains appropriate.

4.4 Negative Binomial

```
Call:
glm.nb(formula = Total.Number.of.Family.members ~ log_income +
    log_food_exp + Household.Head.Age + log_floor_area + House.Age +
    Number.of.bedrooms + Household.Head.Sex + Type.of.Household +
    Electricity, data = df_clean, init.theta = 120066.8686, link = log)

Coefficients:
                                                        Estimate Std. Error
(Intercept)                                           -2.1341939  0.2579964
```

```
log_income                                            -0.3803536  0.0369116
log_food_exp                                           0.7577639  0.0417809
Household.Head.Age                                    -0.0023998  0.0009318
log_floor_area                                         0.0076786  0.0200145
House.Age                                             -0.0032621  0.0010749
Number.of.bedrooms                                     0.0053247  0.0142699
Household.Head.SexMale                                 0.2015365  0.0314009
Type.of.HouseholdSingle Family                        -0.2923885  0.0254673
Type.of.HouseholdTwo or More Nonrelated Persons/Members  0.0128224  0.1207501
ElectricityYes                                        -0.0389571  0.0324910
                                                      z value Pr(>|z|)
(Intercept)                                            -8.272  < 2e-16 ***
log_income                                            -10.304  < 2e-16 ***
log_food_exp                                           18.137  < 2e-16 ***
Household.Head.Age                                     -2.575  0.01001 *
log_floor_area                                          0.384  0.70124
House.Age                                              -3.035  0.00241 **
Number.of.bedrooms                                      0.373  0.70905
Household.Head.SexMale                                  6.418 1.38e-10 ***
Type.of.HouseholdSingle Family                        -11.481  < 2e-16 ***
Type.of.HouseholdTwo or More Nonrelated Persons/Members  0.106  0.91543
ElectricityYes                                         -1.199  0.23052
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(120066.9) family taken to be 1)

    Null deviance: 1854.6  on 1787  degrees of freedom
Residual deviance: 1039.9  on 1777  degrees of freedom
AIC: 6919.3

Number of Fisher Scoring iterations: 1

              Theta:  120067
          Std. Err.:  413888
Warning while fitting theta: iteration limit reached

 2 x log-likelihood:  -6895.34

  =  120066.9
```

The estimated dispersion parameter in the negative binomial model ($\theta = 120066.9$) is extremely large, indicating that the model effectively reduces to a Poisson model. This suggests

no evidence of overdispersion. Additionally, the AIC of the negative binomial model was slightly higher than that of the Poisson model, supporting the use of the simpler Poisson specification.

4.5 Main Findings

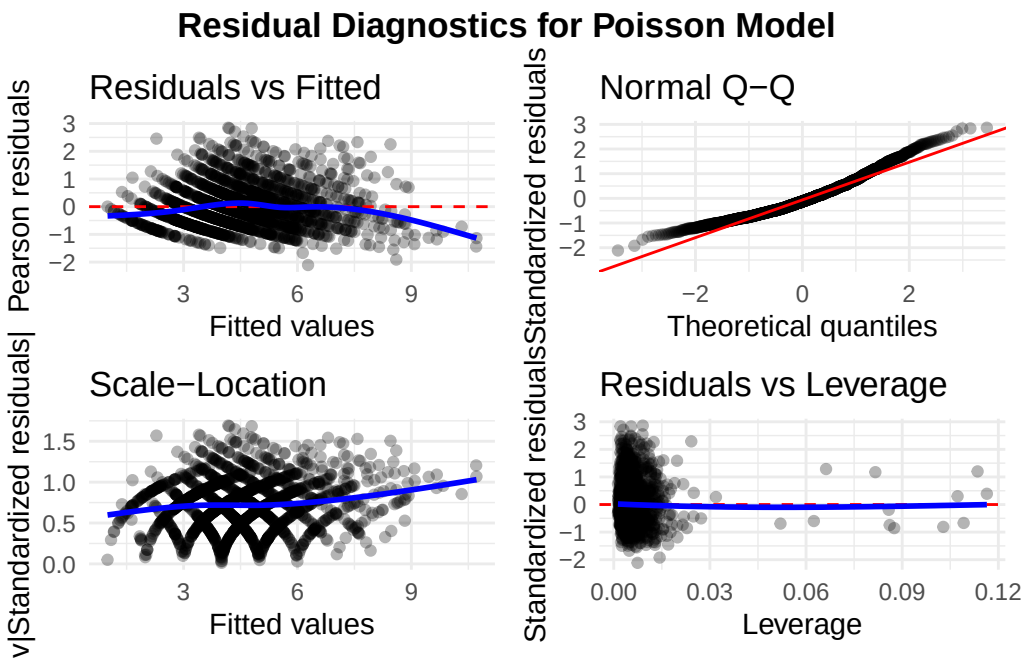
The main findings indicate that household size is most strongly related to food expenditure, income, household head sex, and household type. Higher food expenditure is linked to larger households, whereas higher income is associated with smaller households. Male-headed households tend to have larger expected household sizes, while single-family households tend to have smaller expected household sizes compared with the reference group. Apart from that, household head age and house age have small but statistically significant negative effects.

5 Model Comparison

Model comparison confirmed that the Poisson model is appropriate. The Poisson and Negative Binomial models produced almost identical coefficient estimates, standard errors, and significance patterns, indicating that the substantive findings are highly consistent across specifications. The Poisson model also achieved a slightly lower AIC (6917.3) than the Negative Binomial model (6919.3). Therefore, the Negative Binomial model did not improve model fit, and the simpler Poisson model is preferred.

6 Diagnostics

6.1 Residual Diagnostics



A set of four diagnostic plots was used to evaluate the adequacy of the Poisson model. The residuals show no clear pattern against fitted values, and the blue smoothing line remains close to zero across the range. The Q-Q plot indicates that the standardized residuals follow the diagonal line closely, with only minor deviations in the tails, suggesting that the residuals are approximately normally distributed. The scale-location plot shows a relatively constant spread of residuals across fitted values, with no obvious funnel shape, indicating that the variance assumption is reasonably satisfied. The residuals vs leverage plot reveals no points in the upper-right or lower-right corners, confirming that there are no influential observations with high leverage and large residuals simultaneously. Overall, these diagnostic plots confirm that the Poisson model provides an adequate fit to the data, with no major violations of model assumptions.

6.2 Multicollinearity

Table 2: Variance Inflation Factor (VIF)

Term	VIF

log_income	5.01
log_food_exp	4.28
Number.of.bedrooms	1.69
log_floor_area	1.57
Household.Head.Age	1.29
ElectricityYes	1.25
Type.of.HouseholdSingle Family	1.21
House.Age	1.14
Household.Head.SexMale	1.12
Type.of.HouseholdTwo or More Nonrelated Persons/Members	1.02

To assess multicollinearity among predictors, Variance Inflation Factor (VIF) was calculated. Most VIF values are below 5, indicating no severe multicollinearity for the majority of predictors. However, `log_income` has a VIF of 5.01, which slightly exceeds the conventional threshold, and is highly correlated with `log_food_exp` (correlation 0.8).

While these values do not indicate extreme multicollinearity, they suggest a moderate degree of dependence between key explanatory variables. This is further supported by the robustness check, where the coefficient of `log_income` changes sign when `log_food_exp` is excluded.

This pattern is consistent with a suppression effect, where correlated variables influence each other's estimated coefficients. Therefore, although multicollinearity is not severe, it may affect the stability and interpretation of individual coefficients.

6.3 Robustness Check: Excluding Food Expenditure

Table 3: Sensitivity Check: Coefficients With and Without Food Expenditure

Variable	With Food	Without Food	Change	Sign Change
log_income	-0.380	0.162	0.542	Yes
Household.Head.Age	-0.002	-0.006	-0.004	No
log_floor_area	0.008	-0.012	-0.020	Yes

The robustness of the findings was assessed by estimating an alternative specification that excludes `log_food_exp`. The table below compares the coefficients across the two models.

The results show that the coefficient for `log_income` changes substantially from negative (-0.380) to positive (0.162), indicating substantial sensitivity to model specification. This suggests that food expenditure is closely related to income and captures an important aspect of household consumption associated with household size.

In contrast, the coefficients for `Household.Head.Age` and `log_floor_area` remain relatively small and show only minor changes across specifications, indicating greater stability.

Overall, the results show that most coefficients are stable, but the effect of income is sensitive to model specification.

7 Conclusion

In conclusion, this study investigated the factors associated with household size in the Philippines using data from 1,788 households. The results show that household size is most strongly related to food expenditure, household type, income, and household head sex. In particular, higher food expenditure is associated with larger household size, while single-family households tend to have fewer members than the reference household type. Income shows a weak negative association in the full model; however, this relationship changes when food expenditure is excluded, suggesting that food expenditure may partly account for the association between income and household size. Male-headed households also tend to have slightly larger family sizes than female-headed households.

Diagnostic checks indicated that the Poisson model was adequate, with no evidence of overdispersion and slight underdispersion. However, the deviation was not substantial, and the Poisson specification remained appropriate. Comparison with the Negative Binomial model further supported this conclusion.

Overall, the model provides a useful baseline for understanding household size determinants in this population, although future work could explore regional variation and possible interaction effects.